CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
Higher 1

CANDIDATE NAME			
CLASS	**2T**	**INDEX NUMBER**	

BIOLOGY

8875/02

Paper 2 Core Paper

Wednesday 2 September 2009
2 hours

Additional materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and HT group on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

Section A Structured Questions

Answer **all** questions.
Write your answers in the spaces provided on the question paper.
Show All working

Section B Essay Questions

Answer **one** question.
Write your answers on the separate answer paper.
All working for numerical answers must be shown.

At the end of the examination,
1. fasten all your work securely together;
2. enter the number of the Section B question you have answered in the grid opposite.

The number of marks is given in brackets [ ] at the end of each question or part question.

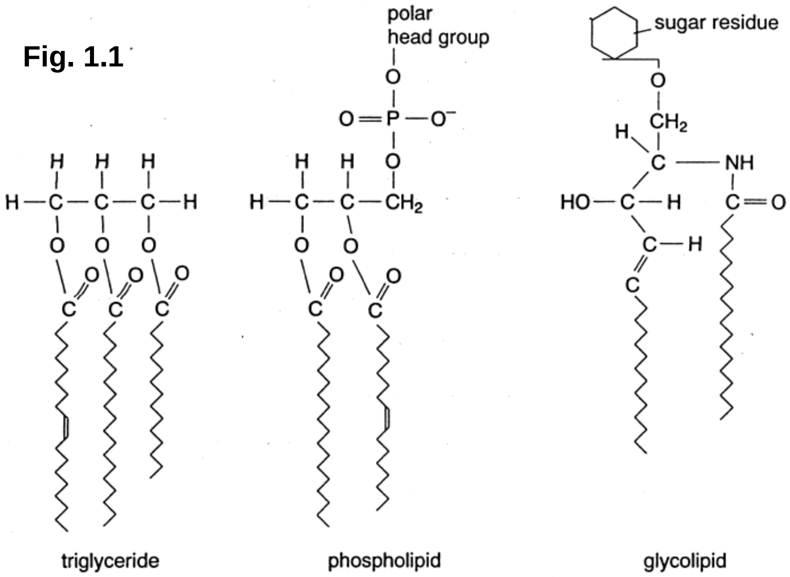
For Examiner's Use	
Section A	**40**
1 [6]	
2 [9]	
3 [9]	
4 [7]	
5 [9]	
Section B	**20**
6 or 7	

Section A

Answer all questions in this section

1 Fig. 1.1 shows the molecular structure of a triglyceride, phospholipid and glycolipid.

Fig. 1.1



(a) Identify the main structural feature common to all three molecules. [1]

Hydrocarbon/fatty acid chains or **Hydrophobic tails**.

(b) Complete the table below for each molecule, describing their differing roles in cells in relation to their differences in molecular structure. [4]

	Structure	**Role in Cells**
Triglyceride	**3 fatty acid chains** linked to **one glycerol molecule** through **ester linkages**. does not contain polar groups	As a **respiratory substrate** for **energy** / in **storage** / **insulation**.
Phospholipid	**Hydrophilic head**: Made up of glycerol, phosphate group and choline group. Accept phosphate / polar head **Hydrophobic**: 2 fatty acid chains	Used in **membrane structure**, forms the lipid bilayers of membranes.
Glycolipid	**Sugar / carbohydrate residue** **2 fatty acid chains**	**Electrical insulation** **cell recognition** / helps cells bind.

(c) State the role of cholesterol in cell membranes. [1]

1. **Maintain membrane fluidity** when temperature changes / **Provides mechanical stability**.
2. **Preventing leakage** of polar molecules.

[Total: 6]

2 In *Silene alba* (White Campion), there are separate male and female plants. Male plants are XY and female plants are XX. In 1912, Baur described the first example of sex-linkage in plants. He discovered an unusual form of *Silene alba* with narrow leaves. It was found to be a male plant. Pollen from this plant was used to fertilise a female plant with normal broad leaves, and all progeny, male and female, had normal broad leaves.

When these normal broad-leaved plants were cross fertilised, they produced 167 plants with broad leaves and 60 with narrow leaves. When the narrow-leaved plants were grown to maturity, all were male, while of the broadleaved plants, 56 were male and 111 were female.

(a) Draw a genetic diagram to explain both crosses. [6]

Symbols used

Let B: allele for broad leaves   b: allele for narrow leaves
X: represents X chromosome   Y: represents Y chromosome
1 mark

Cross 1

Parental phenotype: narrow leaves male X broad leaves female

Parental genotype: X^bY X^BX^B

Gametes: X^b Y X^B X^B

1 mark

F_1 genotype:

♂ \ ♀	X^B	X^B
X^b	X^BX^b	X^BX^b
Y	X^BY	X^BY

F_1 phenotypic ratio: All broad leaves.

1 mark

Cross 2

F_1 phenotype: broad leaves male X broad leaves female

F_1 genotype: X^BY X^BX^b

Gametes: X^B Y X^B X^b

1 mark

F_2 genotype:

♂ \ ♀	X^B	X^b
X^B	X^BX^B	X^BX^b
Y	X^BY	X^bY

1 mark

F_2 phenotypic ratio: 3 broad leaves plant : 1 narrow leaves plant
All female plants have broad leaves
1 broad leaves male : 1 narrow leaves male

1 mark

(b) Explain fully why there were no female narrow-leaved plants. [2]

1. No female narrow-leaved plants due to the **absence of double recessive or X^bX^b individuals**.

2. Male plant has broad leaves, thus, all female plants would **have a copy of the dominant allele** on the X chromosome **inherited from the male plant**. Hence, all females have broad leaves.

(c) Suggest how a female narrow-leaved plant could be produced. [1]

1. Crossing **a narrow-leaved male plant** with **a carrier / heterozygous female** (X^bY cross with X^BX^b).
2. Cross **a male plant with narrow leaves from F_2** with the **female plants with broad leaves from F_2** as well.

[Total: 9]

3 Fig. 3.1 shows a diagram of a ribosome synthesising a protein - translation.

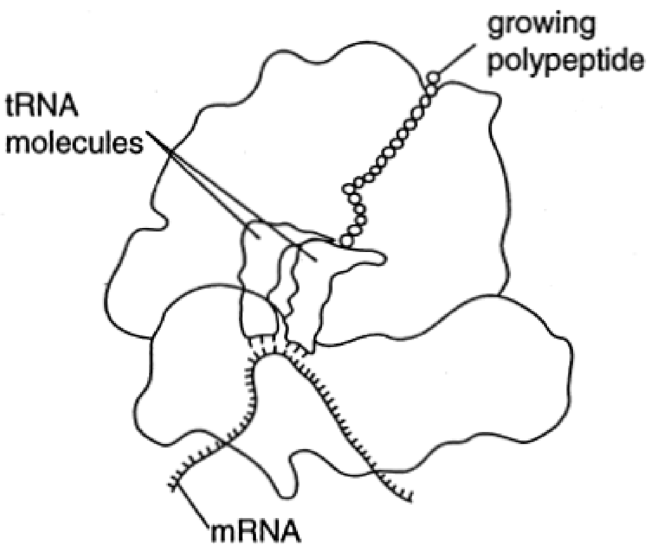


Fig. 3.1

(a) Explain how the tRNA binds to the mRNA. [2]

1. **Complementary base pairing** between the **codon on the mRNA** and the **tRNA anticodon**.
2. **A-U** and **G-C**, **hydrogen bonding**.

(b) List three ways in which transcription differs from translation. [3]

1. Transcription occurring in the **nucleus**, with translation taking place in the **cytoplasm/ribosomes**.

2. Synthesis of mRNA from **DNA template** and protein from **mRNA**.

3. Transcription linking nucleotides through **phosphodiester bonds**. Translation linking amino acids through **peptide bonds**.

(c) Describe the main features of ribosome structure. [3]

1. A ribosome is made up of two subunits, **large** and **small subunits**.
   (Prokaryotes 70S ribosome: 30S small subunit, 50S large subunit; Eukaryotes 80S ribosomes: 40S small subunit, 60S large subunit)
2. Ribosomal subunits are made up of **ribosomal RNA** and **protein**.
3. Size of 20 – 30 nm.
4. Each ribosome has three binding sites for tRNA, the **P site** (peptidyl-tRNA site), the **A site** (aminoacyl-tRNA site) and the **E site** (exit site).

For info:
P site holds the tRNA carrying the growing polypeptide chain.
A site holds the tRNA carrying the next amino acid to be added to the chain.
Discharged tRNAs leave the ribosome from the E site.

(d) State where in the cells ribosomes are made. [1]

1. Nucleolus

[Total: 9]

4 Evolutionary changes in the beaks of Darwin's finches have been instrumental in the adaptive radiation of these birds in the Galapagos Islands.

There are a dozen species that are closely related to one another showing remarkable differences in beak shape and size, and in eating habits. They have all arisen from a common ancestor, but have evolved to exploit different food sources.

The precise dimensions (length depth and width) of each species' beak are crucial to their survival, and fluctuations in the environment lead to selection that changes the relative success of birds with various beak shapes.

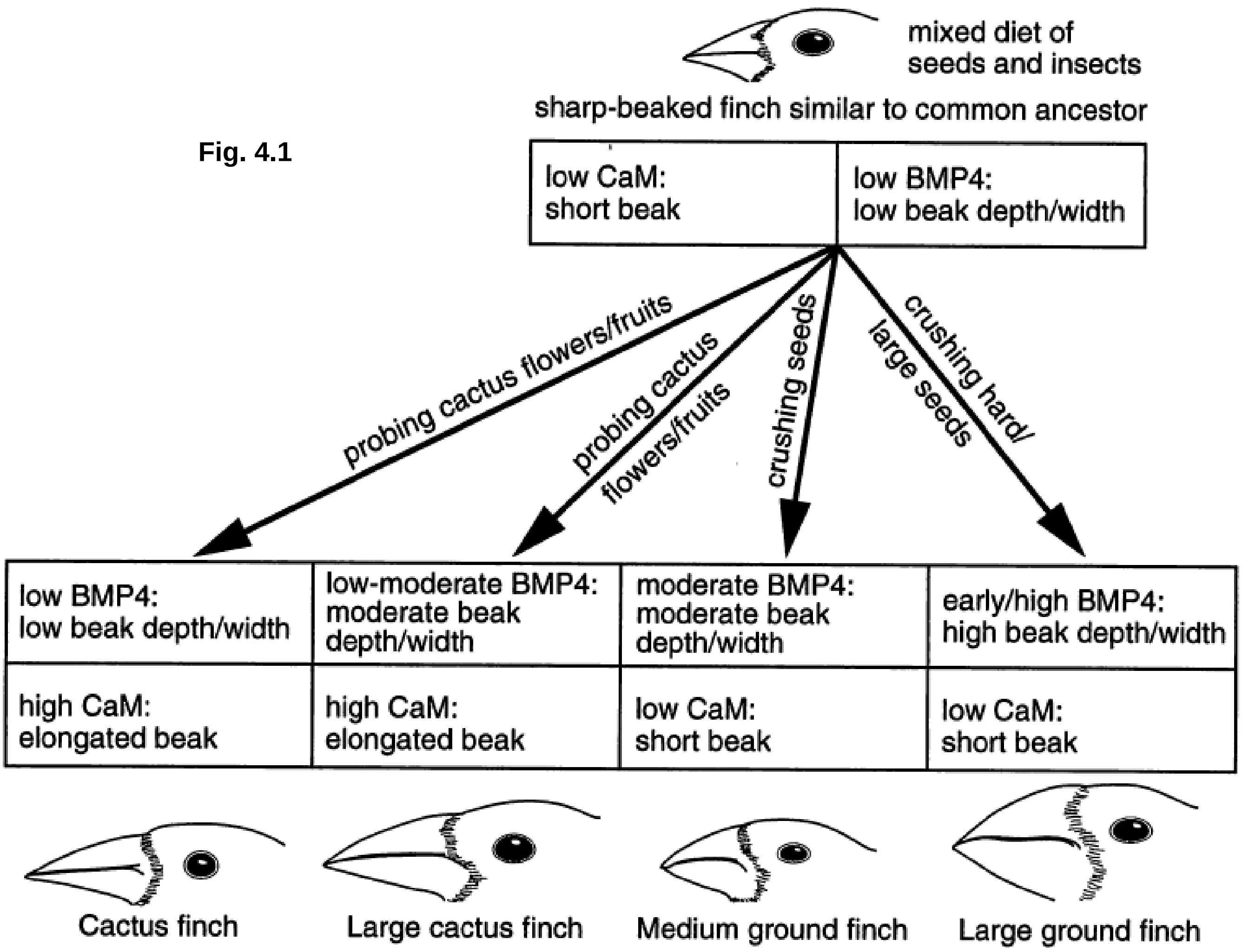
The molecular basis for variation in beak size and shape appears to be due to two genes:

the gene for **bone morphogenetic protein 4** (*BMP4*) that encodes a cell-cell signaling protein that is expressed more during the embryonic development of the deep and wide beaks of ground finches than during the development of narrower beaks.

the gene for **calmodulin** (*CaM*) that encodes a calcium signaling protein that is expressed at much higher levels in the long and pointed beaks of cactus finch embryos.

Fig. 4.1 shows how these two genes interact to produce variation in beak size and shape.

Fig. 4.1



(a) Describe the environmental factors that act as forces of natural selection on finches in the Galapagos Islands. [3]

1. **Type** of food, any two from cactus flowers, fruits, seeds and hard and large seeds.

2. **Availability** of food, **fluctuations in environment** leading to variation in supply of food during different seasons.

3. **Competition** for food, **cactus finch competing with large cactus finch** or **medium ground finch competing with large ground finch**.

(b) Explain how the molecular evidence described supports Darwin's theory of natural selection. [4]

1. Most offspring **die before they reach reproductive age** as a result of **selection pressure** exerted by the environment like type, availability and competition for food.

2. Individuals in a population are **genetically different** from one another, hence there is **variation within a species**.

3. **BMP4 and CaM being present in the ancestor** and that there are different levels of gene expression which in turn leads to **variation**.

4. Finches that can feed with the features/characteristics **best suited for the environment** (or genotypes that express phenotypes) are **more likely to survive** to **reproductive age**.

5. Successful finches **produce offspring** with characteristics or features similar to themselves as the offspring **inherit the favourable genes** from the parents.

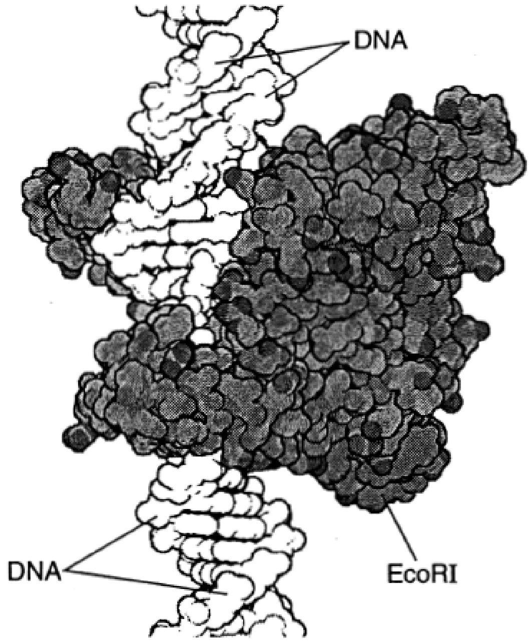
6. Hence the number of individuals with favourable genes will **predominate in a population** / the **frequency of certain** alleles /genotypes will **increase**.

(any 4 points)

[Total: 7]

5 Fig. 5.1 shows the restriction enzyme EcoRI, extracted from the bacterium *Escherichia coli*, attached to a DNA molecule.

Fig. 5.1



(a) Outline how restriction enzymes may be used in the formation of recombinant DNA. [4]

1. **Cutting** both strands of the **source DNA** and the **vector DNA** at precise points **within a restriction site**.
2. The **sugar-phosphate backbones** in the two DNA strands are cleaved in a staggered manner via **hydrolysis** or **breaking of phosphodiester bonds**.
3. Forming a **single-stranded end** called "**sticky end**" or "**blunt end**".
4. These "sticky ends" form hydrogen bonds with **complementary bases** on other DNA molecules cut with the same restriction enzyme.

(b) Suggest an advantage to bacteria in having restriction enzymes in the cytoplasm of their cells. [1]

1. Enzymes would cut and therefore **destroy any foreign DNA** such as that from invading viruses.

(c) Name two **other** enzymes used in the formation of recombinant DNA and briefly describe their role. [4]

1. **DNA ligase**, catalyse the formation **covalent phosphodiester bonds** that close up the sugar-phosphate **backbones of DNA** strands.

2. **Reverse transcriptase**, transcribes **an RNA template into DNA**, providing an RNA to DNA information flow so that **complementary DNA** is obtained from **mRNA** isolated.

3. Or Reverse transcriptase / Terminal transferases with corresponding function.

[Total: 9]

Section B

Answer EITHER 6 OR 7 on the Answer Paper provided.

6 **(a)** Explain the unique features of stem cells and the normal functions of stem cells in a living organism, using appropriate examples to illustrate. [15]

1. **Unspecialized**. There is an **absence of tissue-specific structures** that allow it to perform specialized functions.
2. **Able to proliferate**. Capable of **continually renewing** and **dividing** (through cell division) for long periods.
3. Stem cells are able to **proliferate** (replicate many times), unlike normal cells.
4. **Capable of differentiating into specialized cells types under appropriate conditions.**
5. Stem cells are able to undergo **differentiation** (give rise to specialized cells), when triggered by **intracellular** and **extracellular** signals.
6. The **intracellular signals** are controlled by the **cell's genes** which are interspersed along long strands of DNA, which carries coded instructions for all the structures and functions of a cell.
7. The **extracellular signals** are **chemicals** secreted by **other cells**, physical **contact** with neighbouring cell and **certain molecules** and **compounds surrounding a cell** in an organism.

(Must have points 1, 2, 3 and 4, Max 6)

1. **Totipotent cells** are cells that retains the **complete set of genetic information** and have the potential to become **any cell type** in the adult body, **any cell** of the **extraembryonic membranes**
1. Example of such is the **fertilised egg** and the **first 4 cells** produced by its cleavage.
3. **Pluripotent stem cells** are cells that retains the complete set of genetic information have the potential to become any cell type in the adult body but not those of the extraembryonic membranes.
4. These cells are usually isolated from embryonic tissue and can be grown in culture, but require special methods to prevent them from differentiating.
5. E.g. Embryonic Stem (ES) Cells. These are isolated from the inner cell mass of the blastocyst.
6. **Multipotent stem cells** are cells that retains the complete set of genetic information have the potential to differentiate Into a limited number of cell types, are usually used to replace dead or damaged cells.
7. These cells are usually found in most organs of the body and if accumulates sufficient mutations may produce a clone of cancer cells.
8. Adult stem cell is an undifferentiated cell found among differentiated cells in a tissue or organ, which can renew and differentiate to yield major specialised cell types of tissue or organ.
9. They exist in a very small number in each tissue and remain non-dividing for many years until they are activated by disease or tissue injury
10. Then they enter normal differentiation pathways to form specialized cell types of the tissue in which they reside.
11. Therefore, the primary role of adult stem cells in living organism is to maintain and repair the tissue in which they are found.
12. Examples of such are: Hematopoietic stem cells which give rise to all the types of blood cells: erythrocytes, B lymphocytes, T lymphocytes, neutrophils, basophils, eosinophils, monocytes, macrophages, and platelets.
13. Bone marrow stromal cells which give rise to a variety of cell types: bone cells (osteocytes), cartilage cells (chondrocytes), fat cells (adipocytes), and other kinds of connective tissue cells such as those in tendons.
14. Neural stem cells in the brain which give rise to its three major cell types: nerve cells (neurons) and two categories of non-neuronal cells—astrocytes and oligodendrocytes.
15. Epithelial stem cells in the lining of the digestive tract give rise to several cell types: absorptive cells, goblet cells, Paneth cells, and enteroendocrine cells.
16. Skin stem cells occur in the basal layer of the epidermis and at the base of hair follicles. The epidermal stem cells give rise to keratinocytes, which migrate to the surface of the skin and form a protective layer. The follicular stem cells can give rise to both the hair follicle and to the epidermis.

(Must have points 1, 3 and 6, Max 12)

6 **(b)** Explain, with examples, what is meant by the terms gene mutation and chromosome aberration. [5]

1. Gene mutation is a **change in the structure of DNA** at a **single locus** while **chromosome mutation** is the **change** in the **number** or **structure** of **chromosomes**.
2. Gene mutation could take the forms of duplication, insertion of **DNA** at a **single locus, insertion, deletion, inversion or substitution of** new **bases**.
3. **Chromosomal mutations** could arise from **structural changes in chromosomes** such as **inversion**, **translocation** and **deletion** and **duplication** of a **region** of **chromosome**.
4. **Chromosomal mutations** involve the **loss** or **gain** of **single chromosomes**, **aneuploidy** occurs when **pair chromosomes fail to separate** during anaphase I of meiosis, giving rise to **non-disjunction.**
5. **Chromosomal mutations** also involve an **increase of entire haploids sets** of chromosomes, **polyploidy**.
6. **Polyploidy in plants** result from **autopolypoidy** (where the number of chromosomes within a plant increase) and **alloploidy** (where the chromosome number in sterile hybrids become sterile and produce fertile hybrids)

[Total: 20]

7 **(a)** Compare the structure and organization of prokaryotic and eukaryotic genomes. [12]

On level of the overall **Structure of Genome**

1. Prokaryotes have **no nuclear membrane**, genome exist as **a nucleoid** / simultaneous transcription and translation can occur / fewer levels of control of gene expression.
   Eukaryotes have **nuclear membrane**, genome is enclosed within it / Absence of simultaneous transcription and translation / more levels of control of gene expression.

2. Prokaryotes have **only one origin of replication** as they have a smaller genome while eukaryotes have **multiple origins of replication** due to more chromosomes present.

On the level of **Linearity and Circularity**

3. DNA in prokaryotes occurs in **a loop**, i.e. circular while eukaryotic DNA is **linear**.

On the level of **Size**

4. Prokaryotes: Smaller genome and smaller number of bases / smaller number of genes / coding regions.
   Eukaryotes: More than one chromosome and larger number of chromosomes and bases / genome / larger number of genes / coding regions.

5. Prokaryotes: Single chromosome, usually haploid (with only 1 set of chromosome) / no pairing of homologous chromosomes possible.
   Eukaryotes: Multiple chromosomes, usually diploid (with 2 sets of chromosomes) / pairing of homologous chromosomes possible / Enables genetic recombination during meiosis.

On the level of **Packing of DNA**

6. **Prokaryotic DNA is naked** while eukaryotic DNA is arranged in **nucleosomes wrapped around histones**.

7. Prokaryotic genome is initially packed to form loop domains and undergoes further DNA supercoiling to form the nucleoid.
   Eukaryotes have an **elaborate, multilevel system** of DNA packing to fit all the DNA into the nucleus in preparation for cell division / 10 nm fiber to 30 nm chromatin fiber or solenoid to looped domain forming 300 nm fiber to metaphase chromosome.

8. The **compacting of DNA in prokaryotes is with the aid of RNA and nucleoid proteins** and these proteins differ from the histone proteins present in eukaryotic nuclei. / Does not achieve same level of condensation complexity as eukaryotes.

   DNA packaging in eukaryotes is with the aid of histone proteins and non-histone scaffolding proteins. / Allows for increased structural complexity, folding to higher degree of condensation. / Rate of transcription is controlled by allowing for increased DNA condensation / conversion between euchromatin and heterochromatin states.

<u>On the level of **Presence of Introns / Telomeres / Centromeres**</u>

9. Prokaryotes **do not have introns**, so there is **no splicing**, hence only one possible protein product per gene.
   Eukaryotes **have introns** which allows **for** alternative **splicing** / Different mRNA molecules are produced from the same primary transcript.
10. Prokaryotes do not have telomeres due to their circular DNA and they do not have centromeres as well.
    Eukaryotes have telomeres protect the ends of chromosomes from degradation and fusion with other chromosomes and they have centromeres for kinetochore formation to anchor spindle fibres so that they could be pulled to the poles.

14. Centromere (as described above in point 10)

<u>On the level of **Type of Regulatory Sequences**</u>

11. Prokaryotes **do not have control elements** while eukaryotes **have control elements** which are grouped into either **silencers** or **enhancers** to enable **better control of gene expression** as transcription of a particular gene can be upregulated or downregulated.

<u>On the level of **Mechanism used to Control Transcription**</u>

12. Basic mechanism of controlling gene expression in bacteria is known as the **operon model where** an **operon is a cluster of genes whose expression is controlled by a single operator.**
    Eukaryotes **have control elements** which are grouped into either **silencers** or **enhancers** to enable **better control of gene expression** as transcription of a particular gene can be upregulated or downregulated. / Each eukaryote gene is transcribed separately, with separate transcriptional controls on each gene.

13. Prokaryotes have **polycistronic genes** / allows coordinated control of metabolic enzymes involved in the same pathway / all products of operon are produced in the same amount.

    Eukaryotes have **monocistronic genes** where individual enzymes are controlled individually by separate promoters / different enzymes can be produced in different amounts or not produced at all.

(Any 12 points, must have comparison and comparison must be done at the same level otherwise no marks awarded)

7 **(b)** Explain how human insulin can be synthesized by bacteria. [8]

Step A: Isolation of insulin gene

1. Using mRNA as the template, **reverse transcriptase** catalyses the synthesis of a single strand of DNA, complementary DNA (cDNA), which is **complementary to the base sequence of the mRNA** and eventually forming the double-stranded cDNA.

2. **Extra *BamHI* sticky ends** are added to the cDNA using **terminal transferase**.

Step B: Preparation of plasmid and insulin gene

3. Digest plasmid with the **same restriction enzyme** eg. *BamHI* which cuts within the tetracycline resistance gene and creates compatible**/ complementary sticky ends** on the plasmid.

Step C: Formation of recombinant DNA through insertion gene of interest into vector

4. Insulin DNA and plasmid DNA are mixed together with **ligase enzyme** which joins the DNA molecules by covalent bonds, resulting in a mixture of **recombinant DNA molecules.**

5. The human insulin gene would be inserted **into the *BamHI* restriction site** within the **tetracycline resistance gene**.

Step D: Transformation of bacteria

6. Recombinant plasmid is introduced into bacterium eg. *E.coli* by **transformation** where only about 1% of the bacterial cells would take up the DNA and only a small proportion of these would contain the recombinant plasmid.

Step E: Selecting the bacteria containing the recombinant plasmid

7. Bacterial colonies containing **recombinant plasmid can grow on the ampicillin plate but not on the tetracycline plate** using **replica plating**. The bacterial colonies containing the recombinant DNA plasmids can be picked off from the ampicillin plate.

Step F: Identify bacterial colonies with human insulin gene through nucleic acid hybridization

8. A gene probe which contains a sequence **complementary** to part of the insulin gene would **hybridise with complementary DNA sequence** of the insulin gene. The position of DNA-gene probe hybrid can be located by **autoradiography**.

Step G: Manufacture, extraction and purification of insulin

9. The genetically modified bacteria are subsequently cultured on a large scale **in fermenters**.
10. These **bacteria secrete insulin** which is **extracted, purified** and sold for use to people with diabetes.

[Total: 20]

Table of Specs

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2. DNA and Genomics	7-10	3	
3. Organisation and Control of Prokaryotic and Eukaryotic Genome	11-13		7(a)
4. Genetic Basis for Variation	22-24	2	6(b)
5. Cellular Physiology and Biochemistry	14-18		
6. Evolution	19-21	4	
7. Isolating / Cloning and Sequencing DNA	25-27	5	7(b)
8. Applications of Molecular and Cell Biology	28-30		6(a)
Total			